SOFTWARE ENGINEERING

PROJECT REPORT

Project Name: Karigar

A Trusted Skilled Services Platform

Software Requirements, Design and Implementation

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ABSTRACT

The Karigar platform is a mobile-based service system designed to address the inefficiencies and trust deficits in the skilled labor market within DHA Lahore. Existing methods of hiring plumbers and electricians—primarily through informal networks, unverified contacts, or unreliable digital platforms—suffer from inconsistent service quality, lack of transparency, and safety concerns. Karigar introduces a structured, verification-driven solution that connects clients with certified laborers through a localized and efficient digital interface.

The system focuses on two core service categories—plumbing and electrical work—within a tightly defined geographic scope (DHA phases 1–8 and adjacent blocks). This deliberate limitation enables operational precision, faster response times, and stronger community trust. Key features of the platform include verified laborer onboarding through identity and skill validation, real-time service booking, transparent pricing mechanisms, secure payment processing, and a performance-based rating system. Additionally, the platform incorporates usability enhancements such as bilingual support (Urdu and English) and voice-assisted interactions to accommodate varying literacy levels.

From a technical perspective, the system is designed to meet defined performance, security, and scalability benchmarks to ensure reliability in real-world usage. The development approach emphasizes a minimum viable product (MVP) strategy, allowing rapid deployment followed by iterative improvements based on user feedback and operational data.

This document presents a comprehensive overview of the Karigar system, including project planning, requirement specifications, feasibility analysis, and system design models. The objective is to deliver a robust, user-centric platform that not only simplifies access to skilled services but also establishes a trusted digital ecosystem within the local community.

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1. Introduction

1.1 Project Summary

Karigar is a mobile-based service platform designed to connect residents of DHA Lahore with verified and skilled plumbers and electricians. The system addresses persistent challenges in the local service market, including unreliable labor availability, lack of trust, inconsistent service quality, and non-transparent pricing. By leveraging a verification-driven onboarding process and a hyper-local operational model, Karigar aims to deliver fast, reliable, and secure access to essential home maintenance services.

Unlike generalized service platforms, Karigar adopts a focused approach by limiting its operations to DHA Lahore (Phases 1–8 and adjacent blocks) and two high-demand service categories. This strategic constraint enables tighter quality control, improved response times, and stronger community-level trust. The platform integrates core functionalities such as real-time booking, laborer tracking, secure payment processing, and performance-based rating systems to ensure a seamless user experience.

1.2 Purpose of the System

The purpose of the Karigar system is to provide a structured and trustworthy digital solution for hiring skilled labor within a defined urban community. The system is intended to replace informal and unreliable methods of service acquisition with a standardized platform that ensures verified labor, predictable service quality, and transparent transactions.

From a business perspective, the system aims to establish a localized service ecosystem that prioritizes reliability, safety, and operational efficiency. From a user perspective, it seeks to reduce the time, uncertainty, and risk associated with finding and hiring plumbers and electricians, particularly in urgent or emergency situations.

1.3 Scope of the System

The Karigar platform is designed to operate within a clearly defined functional and geographical scope to ensure controlled deployment and high service quality.

In Scope:

- Mobile application for clients and laborers

- Registration and verification of plumbers and electricians
- Service search, booking, and real-time tracking
- Integrated payment processing (cash and digital methods)
- Rating and review system
- Admin panel for system monitoring and management
- Operations limited to DHA Lahore (Phases 1–8 and adjacent blocks)

Out of Scope:

- Services beyond plumbing and electrical work
- Expansion beyond DHA Lahore in the initial phase
- Desktop/web-based client interface (mobile-first approach)
- Advanced AI-based pricing optimization in initial release

1.4 Report Structure

This document is organized to present a comprehensive and structured overview of the Karigar system. It begins with an analysis of existing solutions and identified gaps in the current market, followed by project planning and development strategy. The document then details the system requirements, including both functional and non-functional aspects, and evaluates the feasibility of the proposed solution. Finally, system design models such as use case, data, and interaction diagrams are presented to illustrate the overall architecture and operational flow of the system.

2. Literature Review

2.1 Existing Solutions

The current landscape for hiring skilled labor in urban areas such as DHA Lahore is dominated by a mix of informal and semi-structured methods. Common approaches include personal contacts, neighbor recommendations, WhatsApp groups, and social media platforms. While these methods provide accessibility, they lack consistency, accountability, and reliability.

In addition to informal channels, several digital platforms have emerged to address this gap, including applications such as Mahir Company and Foran Connection. These platforms attempt to digitize the process of connecting clients with service providers by offering booking interfaces, service listings, and basic verification mechanisms. However, their operational models are typically broad in scope, covering multiple cities and service categories, which often leads to diluted service quality and inconsistent user experiences.

2.2 Limitations of Current Systems

Despite the availability of both informal and digital solutions, several critical limitations persist:

- **Lack of Trust and Verification:** Existing platforms often provide minimal or unclear verification of laborers, leading to safety concerns among users. Informal methods rely entirely on word-of-mouth, which is unreliable and unstructured.

- **Inconsistent Service Quality:** There is no standardized mechanism to ensure consistent quality of work. Ratings, where available, are often insufficient or prone to manipulation.

- **Pricing Uncertainty:** Users frequently face unclear or fluctuating pricing, with little transparency before service delivery. This leads to dissatisfaction and mistrust.

- **Delayed Response Times:** In urgent situations, especially emergencies, finding available labor quickly remains a challenge due to lack of real-time availability tracking.

- **Poor Localization:** Most digital platforms operate on a broad geographical scale, failing to optimize for hyper-local needs such as phase-specific availability within DHA.

2.3 Gap Identification

The analysis of existing solutions highlights a clear gap in the market for a **localized, trust-centric, and efficiency-driven platform**. There is a need for a system that prioritizes:

- Verified and accountable labor through structured identity and skill validation

- Transparent pricing mechanisms for common services

- Rapid response times enabled by hyper-local service allocation

- A reliable rating system tied to actual service delivery
- Accessibility for users with varying levels of digital literacy

Karigar is designed to address these gaps by adopting a focused operational model limited to DHA Lahore and a narrow set of high-demand services. By emphasizing verification, localization, and user-centric design, the platform aims to deliver a more reliable and trustworthy alternative to both informal methods and existing digital solutions.

2.4 Research Methodology

To ensure that the Karigar system is grounded in real-world user needs, a structured requirements gathering process was conducted using a combination of qualitative and quantitative techniques.

2.4.1 Structured Interviews

Structured interviews were conducted to gain deep insights into user behavior, pain points, and expectations from both clients and laborers.

- **Participants:** DHA Residents (15), Laborers (10), Commercial Clients (5)
- **Total Interviews:** 30
- **Duration:** 45–60 minutes per session
- **Mode:** In-person and remote (phone/video)
- **Timeline:** 2 weeks

2.4.2 Survey

A survey was conducted to validate interview findings at scale and quantify user preferences.

- **Target Group:** DHA Residents (Phases 1–8)
- **Sample Size:** 100+ responses
- **Distribution Channels:** WhatsApp groups, Facebook communities
- **Tool Used:** Google Forms
- **Duration:** ~5 minutes per response

- **Timeline:** 1 week

2.4.3 Competitor Analysis

Existing service platforms were analyzed to identify functional gaps and usability limitations.

- **Target Applications:** Mahir Company, Foran Connection
- **Method:** Hands-on usage and feature comparison
- **Duration:** 3 days
- **Analysts:** 2 team members

2.4.4 Key Findings and Insights

The research process yielded several critical insights that directly influenced system design decisions.

Table 1. Key User Insights

Area	***Insight***	***Impact on System***
Trust	Users prioritize verified labor over cheaper options	Strong verification system implemented
Response Time	Emergency response is critical	Real-time availability + fast booking
Pricing	Users dislike uncertain pricing	Fixed pricing introduced
Localization	Preference for nearby laborers	Phase-based matching system
Usability	Laborers prefer Urdu and simple UI	Bilingual + voice support added

These insights form the foundation of the Karigar platform's core features and design strategy.

3. Project Management

3.1 Project Planning and Scheduling

The development of the Karigar platform follows a structured yet adaptive project management approach to ensure timely delivery, quality assurance, and alignment with user requirements. The project is divided into clearly defined phases, each with specific objectives, deliverables, and evaluation checkpoints.

The planning emphasizes a **Minimum Viable Product (MVP)** strategy, allowing early deployment of core features followed by iterative enhancements based on real user feedback.

3.1.1 Development Approach

The project adopts an **Agile Development Methodology**, specifically a **Scrum-based approach**, due to the following reasons:

- **Evolving Requirements:** User needs (especially laborers and residents) are dynamic and better refined through continuous feedback.
- **Incremental Delivery:** Core features such as booking and verification can be developed and tested early.
- **Risk Reduction:** Frequent iterations allow early identification and mitigation of technical and operational risks.
- **User-Centric Design:** Continuous involvement of stakeholders ensures alignment with real-world expectations.

Agile Implementation Strategy:

Table 2. Agile Implementation Strategy

Component	***Description***
Sprint Duration	2 Weeks

Sprint Planning	Define features and tasks for each iteration
Daily Standups	Track progress and resolve blockers
Sprint Review	Demonstrate completed features
Sprint Retrospective	Identify improvements for next sprint

3.1.2 Project Phases and Timeline

The project is divided into five major phases, each contributing to the overall system development lifecycle.

Table 3. Project Phases and Deliverables

Phase	*Duration*	*Key Activities*	*Deliverables*
Phase 1: Research & Analysis	2 Weeks	Interviews, surveys, competitor analysis	Requirement insights, initial scope
Phase 2: System Design	2 Weeks	Use case modeling, ERD, architecture planning	Design diagrams, system blueprint
Phase 3: Development (MVP)	4 Weeks	Core feature implementation	Functional prototype
Phase 4: Testing & Validation	2 Weeks	Unit testing, integration testing, user testing	Tested and validated system
Phase 5: Deployment & Refinement	2 Weeks	Deployment, bug fixes, performance tuning	Final system release

3.1.3 Milestones

To ensure progress tracking and accountability, key project milestones are defined as follows:

Table 4. Project Milestones

Milestone	*Description*	*Expected Outcome*
M1: Requirement Finalization	Completion of all system requirements	Approved SRS document
M2: Design Approval	Finalization of all system diagrams	Verified system architecture
M3: MVP Completion	Core system features implemented	Working application prototype
M4: Testing Completion	System tested against requirements	Quality assurance validation
M5: Final Submission	Complete documentation and system	Project delivery

3.1.4 Task Allocation and Responsibility Matrix

To ensure clarity in roles and accountability, responsibilities are distributed among team members using a simplified RACI (Responsible, Accountable, Consulted, Informed) model.

Table 5. Responsibility Matrix (RACI)

Task	*Responsible*	*Accountable*	*Consulted*	*Informed*
Requirement Analysis	Business Analyst	Project Manager	Stakeholders	Team

System Design	System Architect	Project Manager	Developers	Team
Development	Developers	Project Manager	UI/UX Designer	Stakeholders
Testing	QA/Testers	Project Manager	Developers	Team
Deployment	DevOps/Backend	Project Manager	Support Team	Stakeholders

3.1.5 Risk Management

Effective risk management is essential to ensure project stability and success. The following risks have been identified along with mitigation strategies:

Table 6. Risk Analysis

Risk ID	***Risk Description***	***Impact***	***Probability***	***Mitigation Strategy***
R1	Delay in development	High	Medium	Agile sprints with strict deadlines
R2	Payment integration failure	High	Medium	Use multiple payment gateways
R3	Low user adoption	High	Medium	Focus on trust and local marketing
R4	Data security breach	High	Low	Implement encryption and secure APIs
R5	Laborer onboarding issues	Medium	Medium	Simplify registration and provide support

3.1.6 Tools and Technologies

The project utilizes modern tools and technologies to ensure efficiency, scalability, and maintainability.

Table 7. Tools and Technology Stack

Category	***Tools/Technologies***
Frontend	Flutter / React Native
Backend	Node.js / Django
Database	Firebase / PostgreSQL
APIs	Payment Gateway APIs, NADRA Verification APIs
Project Management	Jira / Trello
Version Control	Git / GitHub
Testing	Postman, Unit Testing Frameworks

3.1.7 Key Performance Indicators (KPIs)

To measure project success and system effectiveness, the following KPIs are defined:

Table 8. Success Metrics

KPI	***Target***

Booking Completion Rate	> 90%
Average Response Time	< 30 minutes
User Satisfaction (CSAT)	> 4.5/5
System Uptime	≥ 99.5%
Laborer Retention Rate	> 85%

3.2 Stakeholder Management

Effective stakeholder management ensures alignment between technical development and business objectives.

3.2.1 Stakeholder Register

Table 9. Complete Stakeholder List

Category	***Stakeholder***	***Role and Responsibility***	***Interest Level***	***Engagement Strategy***
TECHNICAL	Lead Developer/CTO	System architecture, AI search, security	High	Weekly technical reviews
	Mobile App Developer (iOS/Android)	Client & laborer app development	High	Daily standups, sprint planning
	Backend Developer	Server, database, payment APIs	High	Technical documentation reviews
	UI/UX Designer	User flows, interface design	High	User testing sessions
	Database Administrator	Data integrity, security, backups	Medium	Monthly security audits

Category	*Stakeholder*	*Role and Responsibility*	*Interest Level*	*Engagement Strategy*
	Data Privacy Officer	PECA 2016 compliance, data protection	Medium	Quarterly compliance reviews
FRONT-END (BUSINESS)	Project Manager (You)	Overall project leadership	High	Daily oversight
	Marketing Lead	Go-to-market strategy, user acquisition	High	Weekly marketing reviews
	Community Operations Manager	Laborer onboarding, community management	High	Daily ground operations
	Legal & Compliance Advisor	Terms of service, labor laws	Medium	Monthly legal reviews
	Call Center/Support Lead	Voice support, dispute handling	Medium	Training sessions
TOOLS/PARTNERS	Payment Gateway Provider	Transaction processing	Medium	Integration testing
	Verification Partner (NADRA/Tasdeeq)	Background checks	High	API integration, SLA agreement
	Cloud Service Provider	Hosting, scalability	Medium	Performance monitoring
	DHA Security Cell	Laborer gate pass coordination	High	Pre-registration liaison
COMMUNITY	DHA Resident Association Leaders	Trust building, adoption	Medium	Monthly coffee meetings
	Local Mosque Committees	Community endorsement	Medium	Introductory presentations
	First 10 Laborers (Founding Members)	Platform feedback, loyalty		

3.2.2 Stakeholder Classification

Stakeholders are categorized based on their level of involvement and influence:

Table 10. Stakeholder Categories

Category	*Stakeholders*
Primary	Clients (Residents), Laborers
Secondary	Project Team, Admin
External	Payment Providers, Verification Authorities, DHA Administration

3.2.3 Power-Interest Analysis

A Power-Interest Grid is used to determine engagement strategies:

Table 11. Power-Interest Matrix

Stakeholder Group	*Power*	*Interest*	*Strategy*
Project Manager	High	High	Manage closely
Clients	Medium	High	Keep satisfied
Laborers	Medium	High	Engage regularly
Payment Providers	High	Medium	Keep satisfied
Community Leaders	Low	Medium	Keep informed

3.2.4 Communication Plan

Table 12. Communication Strategy

Stakeholder	*Communication Method*	*Frequency*
Development Team	Daily Standups	Daily
Project Manager	Progress Reports	Weekly
Stakeholders	Review Meetings	Bi-weekly
Users (Pilot Phase)	Feedback Sessions	After testing

This structured project management approach ensures that the Karigar system is developed in a controlled, efficient, and user-focused manner while maintaining flexibility to adapt to real-world challenges.

4. System Requirements Specification (SRS)

4.1 User Requirements

4.1.1 Client (Resident / Commercial User) Requirements

- **Account Registration & Profile Management**

UR-CLI-01: Client shall be able to register/login using a mobile number with OTP verification.

UR-CLI-02: Client shall be able to create a profile containing name, phone number, and DHA phase/block.

UR-CLI-03: Client shall be able to edit profile information at any time.

UR-CLI-04: Client location shall be auto-detected with manual override option.

- **Service Discovery & Booking**

UR-CLI-05: Client shall be able to view nearby available laborers on a real-time map.

UR-CLI-06: Client shall be able to filter laborers by:

- Service type (Plumber / Electrician)
- Rating
- Availability

UR-CLI-07: Client shall be able to view laborer profiles including:

- Verified badge
- Ratings & reviews
- Number of completed jobs

UR-CLI-08: Client shall be able to request:

- Immediate service
- Scheduled service (date & time)

UR-CLI-09: Client shall be able to add job details using text, images, or voice notes.

UR-CLI-10: Client shall be able to track the laborer's live location after booking confirmation.

- **Payments & Billing**

UR-CLI-11: Client shall be able to pay using:

- Cash
- Debit/Credit Card
- JazzCash

UR-CLI-12: Client payment shall be held securely until job completion confirmation.

UR-CLI-13: Client shall receive a digital receipt after successful payment.

- **Ratings, Feedback & Support**

UR-CLI-14: Client shall be prompted to rate the laborer (1–5 stars) after job completion.

UR-CLI-15: Client shall be able to submit optional written or voice feedback.

UR-CLI-16: Client shall be able to report issues or disputes via in-app support.

- **Notifications & Accessibility**

UR-CLI-17: Client shall receive notifications for booking status changes.

UR-CLI-18: Client shall receive OTP and payment alerts via SMS/in-app notifications.

UR-CLI-19: Client shall be able to switch the app language between Urdu and English.

UR-CLI-20: Client shall be able to complete the first booking within 3 minutes.

4.1.2 Laborer User Requirements

- **Registration & Verification**

UR-LAB-01: Laborer shall be able to register using a mobile number with OTP verification.

UR-LAB-02: Laborer shall be required to upload CNIC front and back images.

UR-LAB-03: Laborer identity shall be verified through NADRA API integration.

UR-LAB-04: Laborer shall select at least one service category (Plumber/Electrician).

UR-LAB-05: Laborer shall receive a verified badge after successful verification approval.

- **Job Handling & Availability**

UR-LAB-06: Laborer shall receive real-time job requests with job details and client location.

UR-LAB-07: Laborer shall be able to accept or reject job requests within a defined time window.

UR-LAB-08: Laborer shall be able to navigate to job location using in-app maps.

UR-LAB-09: Laborer shall update job status as:

- On the Way
- Arrived
- Work Started

- Work Completed

UR-LAB-10: Laborer shall be able to set availability status (Online/Offline).

- **Earnings & Withdrawals**

UR-LAB-11: Laborer shall receive 80% of the total job payment.

UR-LAB-12: Platform shall deduct a fixed 20% service commission.

UR-LAB-13: Laborer shall be able to withdraw earnings via:

- Bank transfer
- JazzCash

UR-LAB-14: Laborer shall receive instant payment confirmation notifications.

- **Training, Usability & Support**

UR-LAB-15: Laborer shall have access to in-app training material relevant to DHA infrastructure.

UR-LAB-16: Laborer shall be able to use the app in Urdu or English.

UR-LAB-17: Laborer shall be able to send and receive voice notes for job communication.

UR-LAB-18: Laborer shall be able to contact customer support during working hours.

4.1.3 Common User Requirements (Client & Laborer)

- **Security & Privacy**

UR-COM-01: All user data shall be encrypted in transit and at rest.

UR-COM-02: Sensitive data (CNIC, phone number) shall be masked in UI displays.

UR-COM-03: In-app calls shall be routed via proxy numbers.

UR-COM-04: Automatic session timeout shall be enforced for inactive users.

- **Compatibility & Performance**

UR-COM-05: Application shall support Android 8.0+ and iOS 13.0+.

UR-COM-06: Core features shall remain usable on low-bandwidth (2G) networks.

- **Legal & Compliance**

UR-COM-07: Users shall explicitly consent to data collection during onboarding.

UR-COM-08: Privacy Policy and Terms & Conditions shall be available in both Urdu and English.

UR-COM-09: System shall comply with PECA 2016 and local data protection laws.

4.1.4 Admin User Requirements

UR-ADM-01: Admin shall be able to access a centralized dashboard showing users, bookings, and revenue.

UR-ADM-02: Admin shall be able to approve, reject, or suspend laborer accounts.

UR-ADM-03: Admin shall be able to monitor all bookings and dispute cases.

UR-ADM-04: Admin shall be able to view audit logs and system health reports.

4.2 System Requirement – Explained

User Requirement (UR-LAB-03): Laborer identity shall be verified through NADRA API integration.

Step 1: Identify the Requirement Type

This requirement involves external API integration, security, data validation, and failure handling, so it will expand into multiple system requirements across functional and non-functional categories.

Step 2: Write the Core Functional System Requirements

SR-LAB-03.1 — CNIC Input Validation The system shall validate that the entered CNIC follows the format XXXXX-XXXXXXX-X (13 digits) before submitting a verification request.

SR-LAB-03.2 — NADRA API Request The system shall send the laborer's CNIC number, full name, and date of birth to the NADRA API via a secure backend server upon registration submission.

SR-LAB-03.3 — NADRA API Response Handling The system shall process the NADRA API response and set the laborer's verification status to one of three states: Pending, Verified, or Failed.

SR-LAB-03.4 — Verified Badge Assignment The system shall assign a verified badge to the laborer's profile only after NADRA verification succeeds and an admin manually approves the account.

SR-LAB-03.5 — Failed Verification Handling The system shall block the laborer's registration and display an appropriate error message if NADRA returns a failed or invalid response.

SR-LAB-03.6 — API Timeout & Retry The system shall automatically retry the NADRA API call up to 3 times in the event of a timeout before marking the verification as failed.

SR-LAB-03.7 — Admin Notification The system shall notify the admin upon each successful or failed NADRA verification attempt for manual review.

Step 3: Write the Security System Requirements

SR-LAB-03.8 — Encrypted Transmission The system shall communicate with the NADRA API exclusively over HTTPS using TLS 1.2 or higher.

SR-LAB-03.9 — CNIC Encryption at Rest The system shall store all CNIC data in the database using AES encryption.

SR-LAB-03.10 — CNIC Masking in UI The system shall display the CNIC in masked format (e.g., 35202-*****-1) across all UI screens.

SR-LAB-03.11 — API Key Security The system shall store the NADRA API key exclusively in backend environment variables and shall never expose it to the client-side or mobile application.

Step 4: Write the Non-Functional System Requirements

SR-LAB-03.12 — Performance The system shall complete the NADRA verification process within 5 seconds under normal network conditions.

SR-LAB-03.13 — Auditability The system shall log all NADRA verification attempts, including timestamp, CNIC reference, and result, for audit purposes.

SR-LAB-03.14 — Scalability The system shall support a minimum of 50 simultaneous NADRA verification requests without degradation in response time.

Step 5: Write the Database Requirement

SR-LAB-03.15 — Verification Record Storage The system shall store the following fields for each verification attempt:

- laborer_id
- cnic_number (encrypted)
- cnic_verified (boolean)
- verification_status (Pending / Verified / Failed)
- verification_timestamp
- nadra_reference_id

Step 6: Final Traceability

System Requirement	Traces To
SR-LAB-03.1 to SR-LAB-03.7	UR-LAB-03 (Functional)
SR-LAB-03.8 to SR-LAB-03.11	UR-LAB-03 + UR-COM-01, UR-COM-02
SR-LAB-03.12 to SR-LAB-03.14	UR-LAB-03 (Non-Functional)
SR-LAB-03.15	UR-LAB-03 (Data)

4.3 Use Cases

Use Case 1: Register User

Table 13.

Field	*Details*
Use Case ID	UC-01

Use Case Name	Register User
Primary Actor	Unregistered User (Client / Laborer)
Secondary Actors	SMS Gateway, NADRA Verification API
Description	This use case allows a new user to register on the Karigar App using their phone number. Clients are registered directly, while laborers must submit CNIC details for verification.
Preconditions	1. User has installed the Karigar App 2. User has a valid phone number 3. Internet connection is available

Use Case 2: Login User

Table 14.

Field	*Details*
Use Case ID	UC-02
Use Case Name	Login User
Primary Actor	Registered User (Client / Laborer)
Secondary Actors	SMS Gateway
Description	This use case allows an existing user to log in to the Karigar App using OTP-based authentication.
Preconditions	1. User is already registered 2. User has access to their registered phone number

Use Case 3: Verify Laborer CNIC

Table 15.

Field	*Details*
Use Case ID	UC-03
Use Case Name	Verify Laborer CNIC
Primary Actor	Laborer
Secondary Actors	NADRA Verification API, System
Description	This use case verifies a laborer's identity by validating their CNIC details through NADRA before allowing them to offer services.

Preconditions	1. Laborer has completed registration 2. CNIC details and images are provided

Use Case 4: Approve Laborer Application

Table 16.

Field	*Details*
Use Case ID	UC-04
Use Case Name	Approve Laborer Application
Primary Actor	Admin
Secondary Actors	System, Laborer
Description	This use case allows the admin to approve or reject laborer applications after successful CNIC verification.
Preconditions	1. Laborer CNIC is verified 2. Admin is logged into the admin panel

Use Case 5: Search Nearby Laborers

Table 17.

Field	*Details*
Use Case ID	UC-05
Use Case Name	Search Nearby Laborers
Primary Actor	Client
Secondary Actors	System (Maps & Location Services)
Description	This use case allows a client to view nearby available plumbers or electricians within DHA Lahore.
Preconditions	1. Client is logged in 2. Location services are enabled

Use Case 6: Book a Service

Table 18.

Field	*Details*
Use Case ID	UC-06
Use Case Name	Book a Service
Primary Actor	Client
Secondary Actors	Laborer, Notification Service
Description	This use case allows a client to book a selected verified laborer by providing job details.
Preconditions	1. Client is logged in 2. At least one laborer is available nearby

Use Case 7: Accept or Reject Job

Table 19.

Field	*Details*
Use Case ID	UC-07
Use Case Name	Accept or Reject Job
Primary Actor	Laborer
Secondary Actors	Client, Notification Service
Description	This use case allows a laborer to accept or reject a job request sent by a client.
Preconditions	1. Laborer is verified and online 2. A booking request has been received

Use Case 8: Complete Job

Table 20.

Field	*Details*
Use Case ID	UC-08
Use Case Name	Complete Job
Primary Actor	Laborer

Secondary Actors	Client, System
Description	This use case allows a laborer to mark a job as completed after finishing the assigned work.
Preconditions	1. Job has been accepted 2. Laborer has reached the client location

Use Case 9: Process Payment

Table 21.

Field	*Details*
Use Case ID	UC-09
Use Case Name	Process Payment
Primary Actor	System
Secondary Actors	Client, Laborer, Payment Gateway
Description	This use case handles payment processing, including commission deduction and laborer earnings.
Preconditions	1. Job is marked as completed 2. Payment method is selected

Use Case 10: Rate Laborer

Table 22.

Field	*Details*
Use Case ID	UC-10
Use Case Name	Rate Laborer
Primary Actor	Client
Secondary Actors	System
Description	This use case allows a client to rate and review a laborer after job completion.
Preconditions	1. Job is completed 2. Client has not rated the job yet

4.4 User Characteristics

Table 23. User Classes and Characteristics

User Class	*Description*	*Technical Proficiency*	*Key Needs*	*Constraints*
Clients (DHA Residents)	Individuals seeking plumbing/electrical services	Medium	Fast booking, trust, transparency	Limited patience for delays
Laborers (Plumbers/Electricians)	Skilled workers providing services	Low to Medium	Easy job access, quick payments	Language barriers, low literacy
Admin	System managers and operators	High	Monitoring, control, reporting	Requires accurate data

Key Observations:

- Majority of laborers prefer **Urdu interface and voice support**
- Clients prioritize **speed, trust, and verified profiles**
- Admin requires **centralized control and auditability**

System Assumptions and Constraints

Assumptions:

- Users have access to a smartphone
- Internet connectivity is available (minimum 2G support for basic features)
- Payment gateways (e.g., JazzCash) are operational
- NADRA verification APIs are accessible

Constraints:

- System initially limited to DHA Lahore (Phases 1–8)
- Only two service categories (plumbing and electrical)
- Mobile-first platform (no desktop interface)
- Compliance with local regulations (PECA 2016)

4.5 Hardware and Software Requirements

4.6 Non-Functional Requirements

This outlines the essential non-functional requirements for the Karigar App platform, focusing only on what is absolutely necessary for a successful launch in the Pakistani market. These 40 requirements across eight categories represent the minimum viable quality standards that must be met to ensure user trust, regulatory compliance, and operational stability.

i. Performance

Table 24. Performance Requirements

ID	*Requirement*	*Target*
NFR-PER-01	App launch time	< 3 seconds
NFR-PER-02	Booking confirmation	< 2 seconds
NFR-PER-03	Payment processing	< 3 seconds
NFR-PER-04	Location detection	< 2 seconds
NFR-PER-05	Concurrent users support	500 at launch, 2000 by Month 8

ii. Reliability

Table 25. Reliability Requirements

ID	*Requirement*	*Target*
NFR-REL-01	System uptime	99.5%
NFR-REL-02	Booking failure rate	< 1%
NFR-REL-03	Payment failure rate	< 0.5%
NFR-REL-04	Crash rate (apps)	< 0.5% of sessions
NFR-REL-05	Daily backup	Required

iii. Security

Table 26. Security Requirements

ID	*Requirement*	*Target*
NFR-SEC-01	Authentication	OTP via SMS
NFR-SEC-02	Data encryption in transit	TLS 1.3
NFR-SEC-03	Data encryption at rest	AES-256 for PII
NFR-SEC-04	CNIC/Phone masking	Show only last 4 digits
NFR-SEC-05	In-app calling	Proxy numbers only
NFR-SEC-06	Payment data	Never store CVV
NFR-SEC-07	Session timeout	15 min (client), 60 min (laborer)
NFR-SEC-08	Role-based access	3 roles: Client, Laborer, Admin

iv. Usability

Table 27. Usability Requirements

ID	*Requirement*	*Target*
NFR-USA-01	Language	Urdu & English toggle
NFR-USA-02	Literacy support	Voice notes for booking
NFR-USA-03	First booking time	< 3 minutes for new user
NFR-USA-04	Laborer app training	< 30 minutes
NFR-USA-05	User satisfaction	CSAT > 4.5/5

v. Compatibility

Table 28. Compatibility Requirements

ID	*Requirement*	*Target*
NFR-COM-01	Android minimum	Android 8.0 (Oreo)
NFR-COM-02	iOS minimum	iOS 13.0
NFR-COM-03	Network minimum	2G support for basic functions
NFR-COM-04	NADRA verification	UAN/SimSim/Tasdeeq integration
NFR-COM-05	Payment gateways	Minimum 2 (cards + JazzCash)

vi. Legal

Table 29. Legal Requirements

ID	*Requirement*	*Target*
NFR-LGL-01	PECA 2016 compliance	Required
NFR-LGL-02	NADRA verification compliance	Required
NFR-LGL-03	Privacy policy	In-app, Urdu + English
NFR-LGL-04	User consent	Explicit opt-in for data
NFR-LGL-05	Tax compliance	FBR integration for withholding tax

vii. Operational

Table 30. Operational Requirements

ID	*Requirement*	*Target*
NFR-OPS-01	Support hours	8 AM - 10 PM, 7 days
NFR-OPS-02	First response time	< 1 hour
NFR-OPS-03	Monitoring	Real-time system health
NFR-OPS-04	Rollback capability	One-click to previous version
NFR-OPS-05	Audit logs	All financial + admin actions logged

viii. Scalability

Table 31. Scalability Requirements

ID	*Requirement*	*Target*
NFR-SCL-01	Horizontal scaling	Add servers without downtime
NFR-SCL-02	Peak load handling	3x normal load without degradation

4.7 Functional Requirements

This outlines the fundamental features required for Karigar App's minimum viable product (MVP). These 34 core requirements represent the essential building blocks needed to launch a functional platform connecting verified laborers with clients in Pakistan.

i. User Registration

Table 32. User Registration Requirements

ID	*Requirement*
FR-REG-01	Login with phone number (OTP)

ID	***Requirement***
FR-REG-02	CNIC verification via NADRA (laborer only)
FR-REG-03	Basic profile: Name, Phone, Location
FR-REG-04	Service category selection (laborer only)

ii. Client App – Booking

Table 33. Client App – Booking Requirements

ID	***Requirement***
FR-CLT-01	View nearby available laborers on map
FR-CLT-02	Filter by service type
FR-CLT-03	View laborer profile + verified badge
FR-CLT-04	Book laborer immediately
FR-CLT-05	Add job details (text + photos)
FR-CLT-06	Track laborer arrival in real-time

iii. Laborer App – Job

Table 34. Laborer App - Job Requirements

ID	***Requirement***
FR-LAB-01	Receive booking requests
FR-LAB-02	Accept/reject jobs
FR-LAB-03	View job location on map
FR-LAB-04	Update status (On Way → Arrived → Started → Completed)

ID	*Requirement*
FR-LAB-05	Set availability (online/offline)

iv. Payments

Table 35. Payments Requirements

ID	*Requirement*
FR-PAY-01	Multiple payments: Cash, Card, JazzCash
FR-PAY-02	80% to laborer / 20% platform fee
FR-PAY-03	Laborer withdrawal to bank/JazzCash
FR-PAY-04	Payment confirmation receipt

v. Rating

Table 36. Rating Requirements

ID	*Requirement*
FR-RAT-01	Client rates laborer (1-5 stars) after job
FR-RAT-02	Average rating displayed on profile

vi. Notifications

Table 37. Notifications Requirements

ID	*Requirement*
FR-NOT-01	OTP for login
FR-NOT-02	New booking request (to laborer)

ID	*Requirement*
FR-NOT-03	Booking confirmed (to client)
FR-NOT-04	Status updates (to client)
FR-NOT-05	Payment received (to laborer)

vii. Verification

Table 38. Verification Requirements

ID	*Requirement*
FR-VER-01	CNIC front/back photo upload
FR-VER-02	NADRA verification API integration
FR-VER-03	Verified badge on profile

viii. Admin Panel

Table 39. Admin Panel Requirements

ID	*Requirement*
FR-ADM-01	Dashboard: users, bookings, revenue
FR-ADM-02	Approve/reject laborer applications
FR-ADM-03	View all bookings with status
FR-ADM-04	Process laborer withdrawals
FR-ADM-05	Block/unblock users

4.8 Hardware Requirements

The system is designed as a mobile-first platform and requires the following hardware components:

Table 40. Hardware Requirements

Component	*Requirement*	*Description*
Client Device	Smartphone	Android or iOS device for accessing the application
Laborer Device	Smartphone	Basic smartphone capable of running the app
Server Infrastructure	Cloud-based servers	Required for hosting backend services and database
Network	Mobile Data/Wi-Fi	Minimum 2G for basic functionality, 4G recommended
GPS Module	Built-in	Required for location tracking and service matching

4.9 Software Requirements

The Karigar system relies on a combination of mobile, backend, and third-party software components.

Table 41. Software Requirements

Category	*Requirement*	*Description*
Mobile OS (Android)	Android 8.0 or above	Ensures compatibility with majority of devices
Mobile OS (iOS)	iOS 13 or above	Supports modern Apple devices
Frontend Framework	Flutter / React Native	Cross-platform mobile development
Backend Framework	Node.js / Django	API handling and business logic
Database	Firebase / PostgreSQL	Data storage and management
APIs	Payment & Verification APIs	JazzCash, Card APIs, NADRA integration
Security Protocols	TLS 1.3, AES-256	Secure data transmission and storage
Development Tools	Git, GitHub, Jira/Trello	Version control and project management
Testing Tools	Postman, Unit Testing Frameworks	API and system testing

4.10 Traceability Matrix

To ensure completeness and alignment, a traceability matrix links functional requirements to system objectives.

Table 42. Requirement Traceability Matrix

Requirement ID	*Objective*	*Module*
FR-REG-01	User onboarding	Authentication
FR-CLT-04	Service booking	Client Module
FR-LAB-02	Job management	Laborer Module
FR-PAY-01	Payment processing	Payment Module
FR-RAT-01	Quality assurance	Feedback System

5. System Analysis

5.1 Feasibility Study

The feasibility study evaluates the practicality of the Karigar system across technical, economic, and operational dimensions. This assessment ensures that the proposed solution is not only technically implementable but also financially viable and adoptable by its intended users.

5.1.1 Technical Feasibility

The technical feasibility of the Karigar platform is high, given the availability of mature technologies, frameworks, and third-party integrations required for implementation.

The system relies on widely adopted mobile development frameworks (e.g., Flutter/React Native) and backend technologies (e.g., Node.js/Django), which are well-supported and scalable. Integration with external services such as payment gateways (JazzCash, card processing) and identity verification systems (NADRA APIs) is achievable through existing APIs.

Additionally, cloud-based infrastructure enables scalable deployment, ensuring that the system can handle increasing user loads without major architectural changes.

Table 43. Technical Feasibility Assessment

Component	*Availability*	*Complexity*	*Risk Level*	*Remarks*
Mobile App Development	High	Medium	Low	Cross-platform frameworks available
Backend Development	High	Medium	Low	Standard REST API architecture
Payment Integration	High	Medium	Medium	Requires secure handling
NADRA Verification	Medium	High	Medium	Dependent on external access
Cloud Infrastructure	High	Low	Low	Easily scalable

Key Conclusion:

The system is **technically feasible**, with manageable risks primarily associated with third-party integrations.

5.1.2 Economic Feasibility

The economic feasibility evaluates whether the expected benefits of the system justify the development and operational costs.

The Karigar platform follows a **low initial investment, high scalability model**, leveraging existing technologies and a focused geographic scope to minimize costs during early stages.

Table 44. Cost Estimation (Initial Phase)

Cost Category	*Estimated Cost (PKR)*
Development (Team Effort)	300,000
Cloud Hosting (Initial)	20,000/month
Payment Gateway Setup	10,000
Marketing & User Acquisition	50,000
Miscellaneous	20,000

Revenue Model:

- Commission-based (20% per transaction)

- Potential future revenue from premium listings or subscriptions

Table 45. Cost-Benefit Analysis

Factor	*Observation*
Initial Investment	Moderate
Revenue Potential	High (scalable with user base)
Break-even Period	Estimated within 6–12 months
Financial Risk	Medium

Key Conclusion:

The system is **economically viable**, with strong long-term revenue potential and controlled initial costs.

5.1.3 Operational Feasibility

Operational feasibility assesses how well the system fits within the real-world environment of its users and whether it will be adopted effectively.

Based on conducted interviews and surveys, there is a clear demand for a **trusted and efficient platform** for hiring skilled labor in DHA Lahore.

Table 46. Operational Feasibility Factors

Factor	*Assessment*	*Justification*
User Acceptance	High	Strong demand for verified labor
Ease of Use	High	Simple UI with Urdu support
Laborer Adaptability	Medium	Requires onboarding and training
Support Requirements	Medium	Customer support needed for disputes
Market Readiness	High	Existing pain points create opportunity

Key Insights:

- Clients are willing to adopt the system if **trust and response time** are ensured
- Laborers require **training and incentives** for platform adoption
- Hyper-local focus increases **community trust and usability**

5.1.4 Legal and Regulatory Feasibility

The system must comply with local laws and regulations, particularly concerning data privacy, digital transactions, and user verification.

Table 47. Legal Compliance Requirements

Regulation	*Requirement*	*Compliance Strategy*
PECA 2016	Data protection and privacy	Secure data handling and encryption
NADRA Policies	Identity verification	Authorized API integration
Financial Regulations	Digital payment compliance	Use certified payment gateways
User Consent Laws	Explicit data usage consent	In-app consent mechanisms

Key Conclusion:

The system is **legally feasible**, provided all integrations and data handling processes adhere to regulatory standards.

5.1.5 Feasibility Summary Matrix

To provide a consolidated view, the feasibility of the system across all dimensions is summarized below:

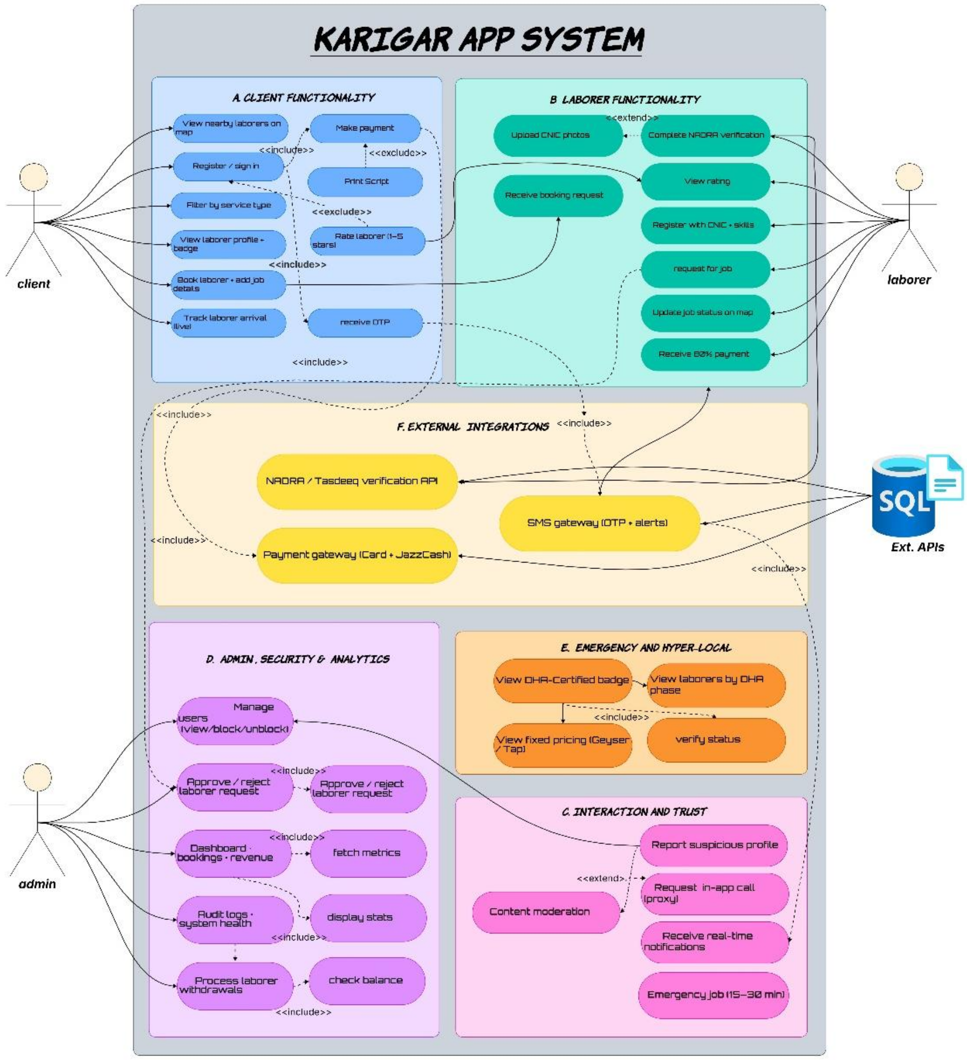
Table 48. Feasibility Summary

Feasibility Type	*Status*	*Risk Level*	*Overall Assessment*
Technical	Feasible	Low–Medium	Strong
Economic	Feasible	Medium	Promising
Operational	Feasible	Medium	High adoption potential
Legal	Feasible	Low	Manageable with compliance

5.2 Functions of the System

5.2.1 Use-Case Diagram

Figure 1. Karigar - Use Case Diagram



Explanation:

The diagram represents the functional architecture and use-case distribution of the *Karigar mobile-based service platform*. It illustrates how different stakeholders—clients, laborers, and administrators—interact with the system, along with supporting external integrations and core service modules.

a. Client Functionality (A)

This module defines the end-to-end journey of a client using the platform. It begins with user onboarding (registration/sign-in) and progresses through service discovery and booking. Clients can filter laborers by service type, view verified profiles, and access rating-based comparisons to ensure informed decision-making.

Once a service is selected, the client can book a laborer, track arrival, and verify the worker via OTP, ensuring security and authenticity. Post-service, the system supports payment processing and feedback submission, creating a closed-loop interaction that feeds into the platform's trust and rating mechanism.

b. Laborer Functionality (B)

This module captures the operational workflow for skilled workers. Laborers undergo profile creation and verification, including CNIC and skill validation, which directly addresses trust issues in the informal labor market.

After onboarding, laborers can receive job requests, update job status in real-time, and manage their availability. The system also enables earnings tracking and digital payment collection, ensuring transparency and financial inclusion. Additionally, performance metrics such as ratings and completed jobs contribute to their credibility within the platform.

c. External Integrations (F)

The system integrates with third-party services to ensure reliability and security:

NADRA/Tasdeeq API for identity verification

Payment gateways (e.g., JazzCash/card systems) for secure transactions

SMS/OTP services for authentication and alerts

These integrations enhance system trustworthiness and automate critical verification and communication processes.

d. Admin, Security & Analytics (D)

The admin module provides centralized control and governance. Administrators can manage users (block/unblock), approve or reject laborer registrations, and monitor system activity.

Advanced features include dashboard analytics (bookings, revenue), system health monitoring, audit logs, and financial oversight (laborer withdrawals). This ensures operational transparency, fraud prevention, and continuous performance evaluation.

e. Emergency & Hyper-Local Services (E)

To align with the DHA-focused scope, this module emphasizes location-based service optimization. Clients can:

- View DHA-certified laborers
- Access phase-specific availability
- Utilize fixed pricing models for common services

This localized approach improves response time and enhances user trust within a defined community.

f. Interaction & Trust Layer (C)

This module strengthens platform reliability and user confidence. It includes:

- Real-time notifications
- In-app communication (e.g., VoIP calls)
- Reporting of suspicious profiles
- Content moderation and emergency job handling

These features collectively ensure safe interactions, rapid communication, and accountability across the platform.

5.2.2 Data Flow Diagram

1. Level 0

a. Client Interaction

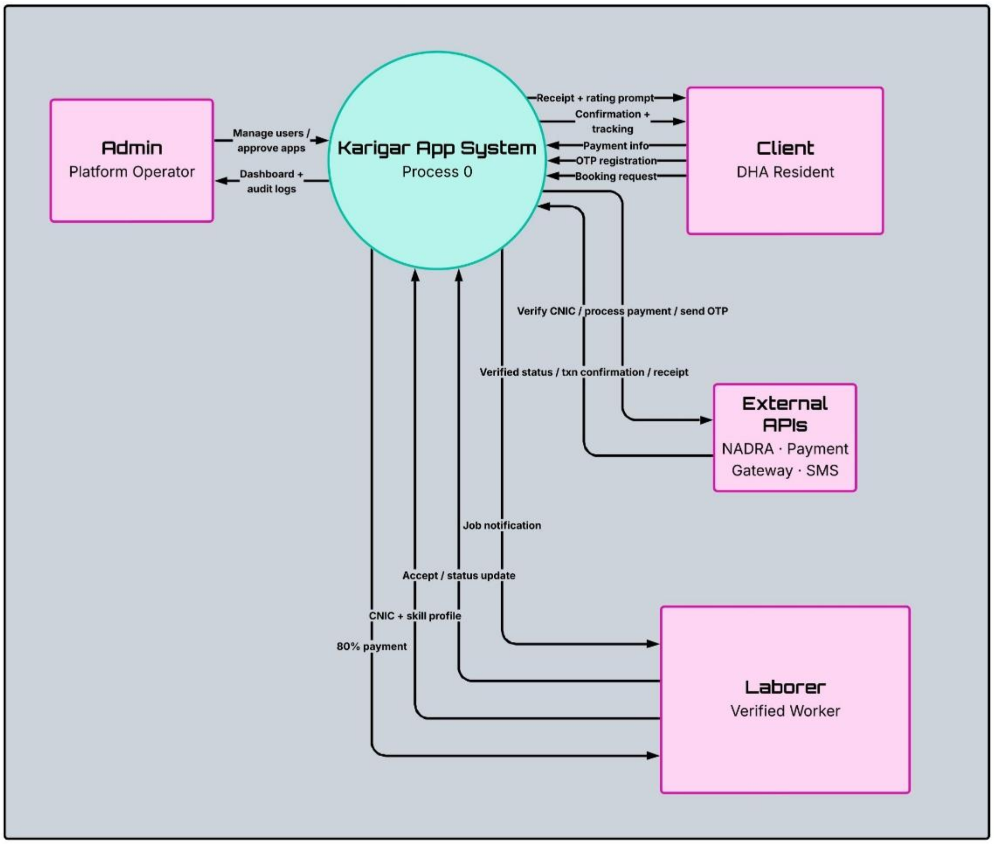
The client (DHA resident) communicates with the system for all service-related actions. Key data flows include:

Inputs to system: registration details, booking requests, and OTP verification

Outputs from system: booking confirmation, tracking updates, payment requests, and service receipt with rating prompt

This flow captures the complete service lifecycle from request initiation to feedback submission.

Figure 2. Karigar - DFD Level 0



b. Laborer Interaction

The laborer (verified worker) provides and receives operational data:

Inputs to system: CNIC details, skill profile, job acceptance, and status updates

Outputs from system: job notifications and payment disbursement (80% share)

This ensures that only verified workers participate and that job execution is tracked in real time.

c. Admin Interaction

The admin (platform operator) oversees system governance:

Inputs to system: user management actions (approve/reject, block/unblock)

Outputs from system: dashboards, analytics, and audit logs

This flow supports monitoring, control, and decision-making at the platform level.

d. External APIs Interaction

The system integrates with third-party services for critical operations:

Outgoing requests: CNIC verification, payment processing, and OTP generation

Incoming responses: verification status, transaction confirmation, and receipt data

These integrations enable secure authentication, financial transactions, and communication.

e. Central Process (Karigar App System)

The central process coordinates all interactions:

Validates users and laborers

Manages booking and job assignment

Handles secure payments and notifications

Maintains system records and transaction history

All data flows pass through this core process, ensuring controlled and structured communication between entities.

2. Level 1

The Level 1 DFD decomposes the main system into multiple sub-processes, showing detailed data movement between processes, data stores, and external entities.

a. Admin Management (P1)

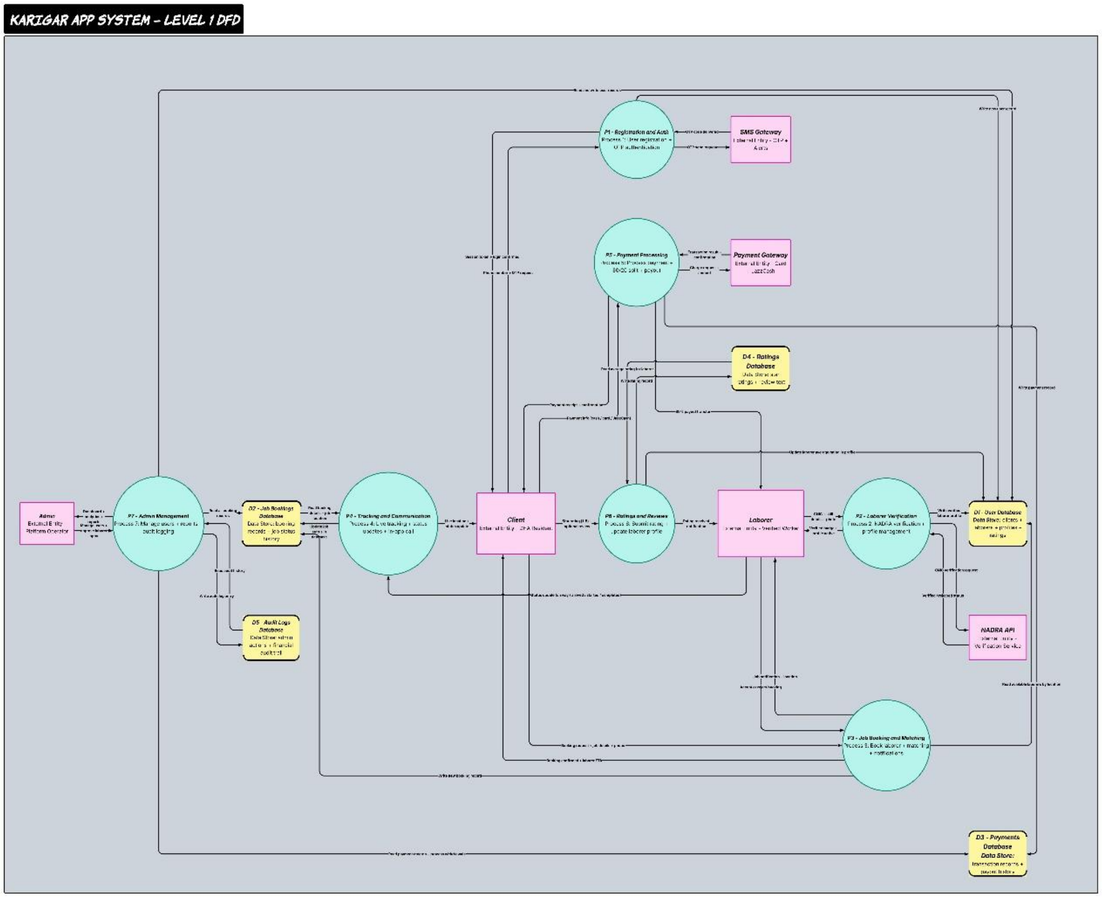
This process handles administrative control over the platform:

Admin sends user approval/rejection, blocking/unblocking requests

System updates records in the Admin Logs Database (D2)

Outputs include system reports, audit logs, and platform status

Figure 3. Karigar - DFD Level 1



b. 2. Job Booking (P2)

This module manages service requests from clients:

Client submits booking request with service details

Data is stored in the Bookings Database (D4)

Relevant job data is forwarded for matching and assignment

c. 3. Tracking & Communication (P3)

This process ensures real-time coordination:

Handles job tracking, notifications, and communication between client and laborer

Uses system data to provide status updates and alerts

Maintains smooth interaction throughout the job lifecycle

d. 4. Ratings & Reviews (P4)

This module manages feedback:

Client submits ratings and reviews after job completion

Data is stored and linked to laborer profiles

Supports transparency and trust within the platform

e. Laborer Verification (P5)

This process validates worker authenticity:

Laborer submits CNIC and skill profile

System interacts with NADRA API for identity verification

Verified data is stored in the User Database (D1)

f. Job Assignment & Matching (P6)

This is the core operational process:

Matches client requests with available laborers

Sends job notifications to laborers

Receives acceptance/rejection and status updates

Updates booking and job progress data

g. Registration & Authentication (P7)

This process handles onboarding and login:

Clients and laborers submit registration/login data

OTP verification is performed via SMS Gateway

Verified user data is stored in the User Database (D1)

h. Payment Processing (P8)

This module manages financial transactions:

Client initiates payment after service completion

System interacts with Payment Gateway

Transaction records are stored in the Payments Database (D3)

Laborer receives 80% payment share, system retains commission

i. 9. Key Data Stores

D1 – User Database: stores client and laborer profiles

D2 – Admin Logs Database: maintains audit and admin activity

D3 – Payments Database: stores transaction records

D4 – Bookings Database: stores job requests and service data

j. 10. External Entities

Client: initiates bookings, payments, and feedback

Laborer: receives jobs, updates status, and earns payments

Admin: monitors and controls the system

External APIs: NADRA (verification), SMS Gateway (OTP), Payment Gateway (transactions)

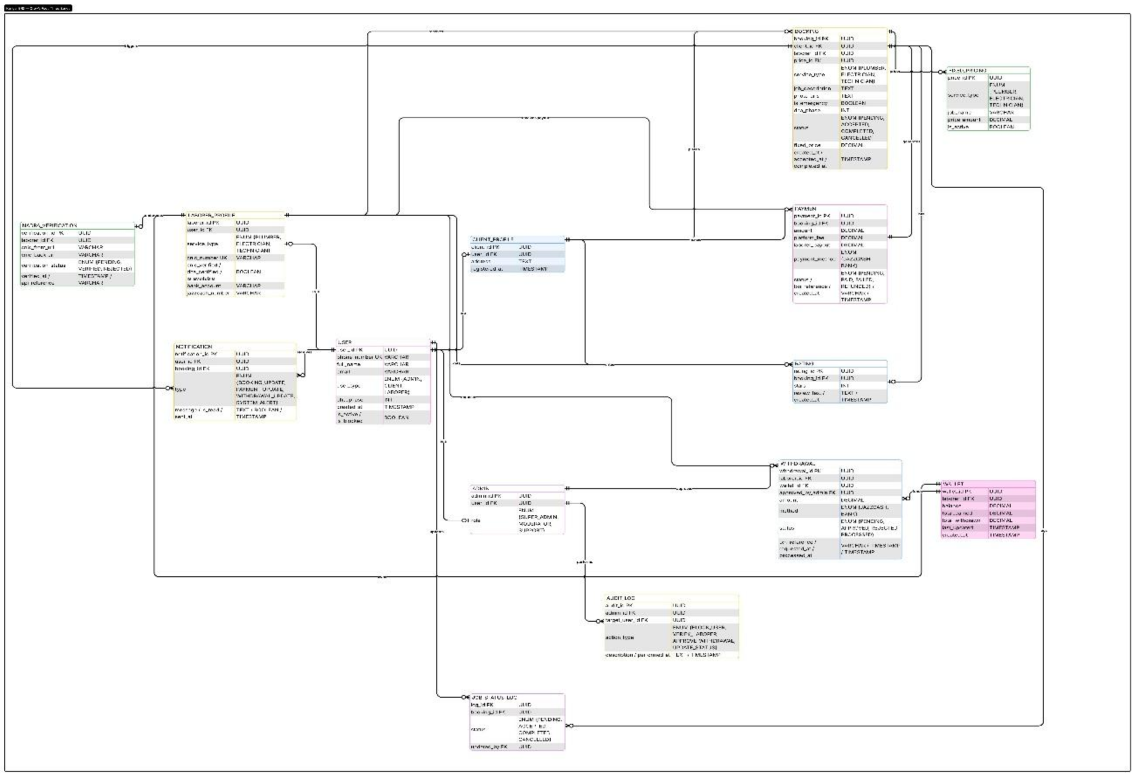
5.3 Data Modeling

5.3.1 ERD Diagram

a. User and Authentication Layer

The schema contains tables dedicated to managing user accounts and access control. This includes identity fields, hashed credentials, contact information, and role-based flags distinguishing clients from laborers. A separate identity verification table stores the documentation and approval status required during the laborer onboarding process, reflecting the platform's emphasis on trust and certification.

Figure 4. Karigar - Entity Relationship Diagram



b. Laborer Profile and Skill Management

A dedicated laborer table extends the base user model with profession-specific attributes, trade category (plumbing or electrical), years of experience, availability status, and geographic zone assignment. Linked skill or certification tables capture validated competencies, ensuring that only verified tradespeople appear in search results.

c. Service Booking and Job Lifecycle

The booking or job table forms the central hub of the schema. It references both the client and the assigned laborer, records the requested service type, timestamps for scheduling and completion, current job status (pending, accepted, in-progress, completed, cancelled), and location metadata. This table drives the primary workflow of the platform.

d. Pricing and Payment Processing

Separate tables manage service pricing structures and transaction records. The pricing table likely stores base rates by trade and service type, while the payments table records transaction IDs, amounts, payment method, status (paid, failed, refunded), and linkage to the relevant booking. This separation supports transparent billing and future integration with payment gateways.

e. Ratings and Reviews

A review or rating table captures post-service feedback submitted by clients. It references the completed booking and the laborer involved, stores a numeric rating and optional comment, and includes a timestamp. This feeds the platform's reputation and accountability mechanism.

f. Notifications

A notifications table manages system-generated and event-driven alerts sent to users, such as booking confirmations, status updates, and payment receipts. It stores the recipient reference, message type, content, read/unread status, and creation time, supporting both push and in-app notification delivery.

g. Audit and Logging

The presence of an audit log table indicates provisions for tracking significant system events, such as account changes, booking modifications, or administrative actions. This is a best-practice inclusion for security compliance, dispute resolution, and system monitoring.

5.3.2 Class Diagram

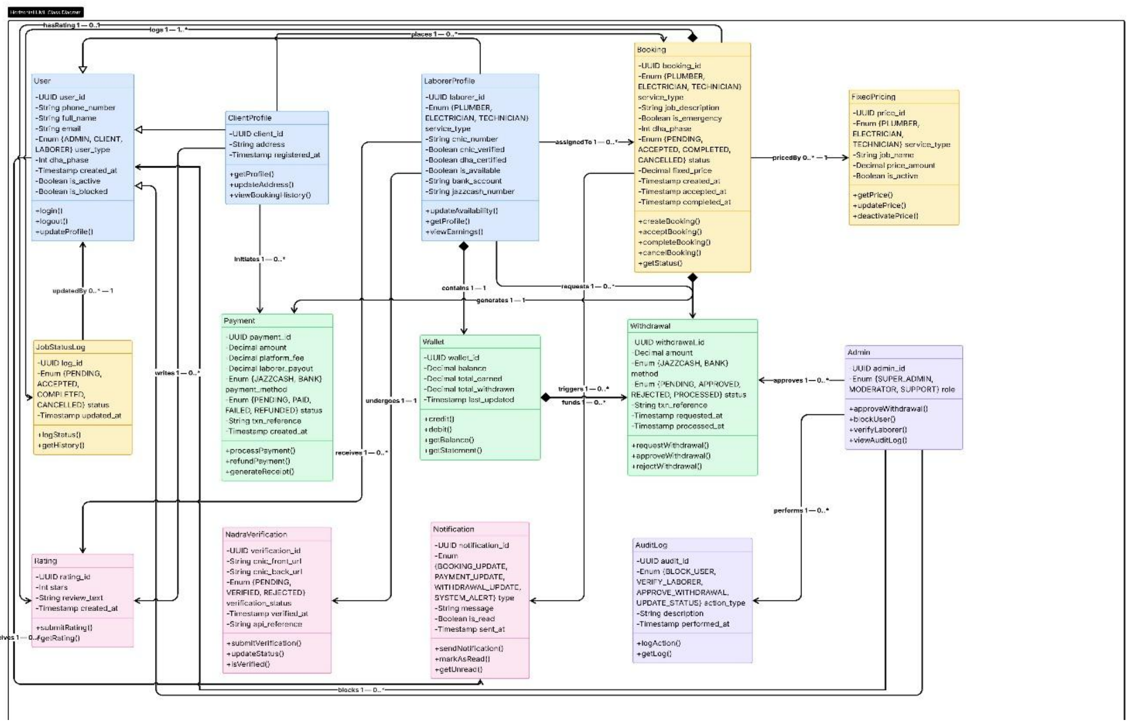
a. User, Base Identity Class

The User class serves as the root identity entity for all platform participants. It holds universal attributes including a UUID-based identifier, phone number, full name, email, and a role enum (ADMIN, CLIENT, LABORER) to distinguish account types. It tracks account lifecycle fields such as created_at, is_active, and is_blocked, and exposes core methods including login(), logout(), and updateProfile(). All role-specific classes associate back to this base.

b. ClientProfile, Client-Side Extension

This class extends the user model for clients, storing a billing address, registration timestamp, and references to booking history. Its methods, updateProfile(), updateAddress(), and viewBookingHistory(), reflect the client's primary interactions with the platform. The association to User is one-to-one, and it initiates bookings in a one-to-many relationship with Booking.

Figure 5. Karigar - Class Diagram



c. LaborerProfile, Service Provider Entity

The LaborerProfile class captures all professional attributes of a laborer: trade type (via PLUMBER, ELECTRICIAN, TECHNICIAN enum), CNIC number, certification verification flags, availability status, average rating, and bank account details for payout. Methods include getProfile(), updateAvailability(), and viewEarnings(). It connects to Booking as the assigned service provider, and links to FixedPricing for its associated service rates.

d. Booking, Central Transaction Class

This is the most critical class in the diagram. It records the full lifecycle of a service request: booking ID, service type enum, location description, emergency flag, status enum (PENDING, ACCEPTED, COMPLETED, CANCELLED), agreed price, and timestamps for creation, acceptance, and completion. It exposes methods for all lifecycle transitions, createBooking(), acceptBooking(), completeBooking(), cancelBooking(), and getStatus(). It maintains associations with ClientProfile, LaborerProfile, Payment, Rating, and Notification.

e. Payment, Financial Transaction Class

The Payment class handles all monetary operations tied to a booking. It stores platform fee, laborer payout, total amount, payment method, transaction reference, and a status enum (PENDING, PAID, FAILED, REFUNDED). Methods include processPayment(), refundPayment(), and generateReceipt(). It associates one-to-one with a completed booking and triggers updates to the Wallet class.

f. Wallet, Laborer Earnings Ledger

Each laborer maintains a Wallet that tracks their current balance, total earned, and total withdrawn amounts, along with the last updated timestamp. Methods getBalance(), debit(), and getStatement() manage the financial state. The wallet is updated upon payment processing and is the source of funds for withdrawal requests.

g. Withdrawal, Payout Request Class

The Withdrawal class manages laborer cash-out requests. It captures the amount, payment method enum (JAZZCASH, BANK), transaction reference, and a status enum (PENDING, APPROVED, REJECTED, PROCESSED). It connects to the Admin class for approval and funds from the Wallet. Methods include requestWithdrawal(), approveWithdrawal(), and reportWithdrawal().

h. FixedPricing, Service Rate Configuration

This class stores the canonical pricing for each service type offered on the platform. It holds the price ID, trade type, job label, price amount, and an is_active flag. Methods getPrice(), updatePrice(), and deactivatePrice() allow dynamic pricing management. It is referenced by Booking at the time of service agreement and linked to laborer profile types.

i. Rating, Feedback and Reputation Class

Post-booking feedback is encapsulated here. The Rating class stores a reference to the booking and laborer, a numeric star rating, an optional review text, and a creation timestamp. Methods submitRating() and getAvgRating() support the platform's reputation system. Its multiplicity allows one booking to generate one rating, while a laborer accumulates many.

j. NadraVerification, Identity Validation Class

This class handles the CNIC-based identity verification workflow for laborer onboarding. It stores the CNIC from/to numbers, a verification status enum (PENDING, VERIFIED, REJECTED), and a sql reference string alongside timestamps. Methods submitVerification(), updateStatus(), and isVerified() drive the administrative approval pipeline.

k. Notification, Messaging and Alert Class

The Notification class manages all system-generated alerts. It stores a notification type enum (BOOKING_UPDATE, PAYMENT_UPDATE, WITHDRAWAL_UPDATE, SYSTEM, ALERT), the message content, a read flag, and a sent timestamp. Methods sendNotification(), markAsRead(), and getUnread() support the real-time communication layer between the system and its users.

l. AuditLog, System Accountability Class

This class records all significant platform actions for traceability. It captures the acting user ID, action type enum (BLOCK_USER, VERIFY_LABORER, APPROVE_WITHDRAWAL, UPDATE_BOOKING), a description string, and a timestamp. The single method getLog() allows administrators to retrieve records. It connects to Admin via a performs relationship.

m. Admin, Platform Governance Class

The Admin class represents platform operators with elevated privileges. It carries a role enum (SUPER_ADMIN, MODERATOR, SUPPORT) and an is_active flag, with methods for approveWithdrawal(), blockUser(), verifyLaborer(), and viewAuditLog(). It approves withdrawals, manages user accounts, and generates audit entries, forming the governance backbone of the platform.

n. JobStatusLog, Booking History Tracker

This supporting class logs each status transition within a booking lifecycle, referencing the booking ID and recording the status enum value with a timestamp. Methods logStatus() and getHistory() provide a traceable history of how a job progressed from creation to resolution, valuable for dispute handling and analytics.

5.3.3 Sequence Diagram

The diagram illustrates the end-to-end interaction flow between five system participants, Client, Client App, Backend, Database, and Laborer App, across four sequential phases.

a. **Login / Authentication:** The client initiates an OTP-based login through the app. The backend sends an OTP request and upon successful verification, establishes an authenticated session.

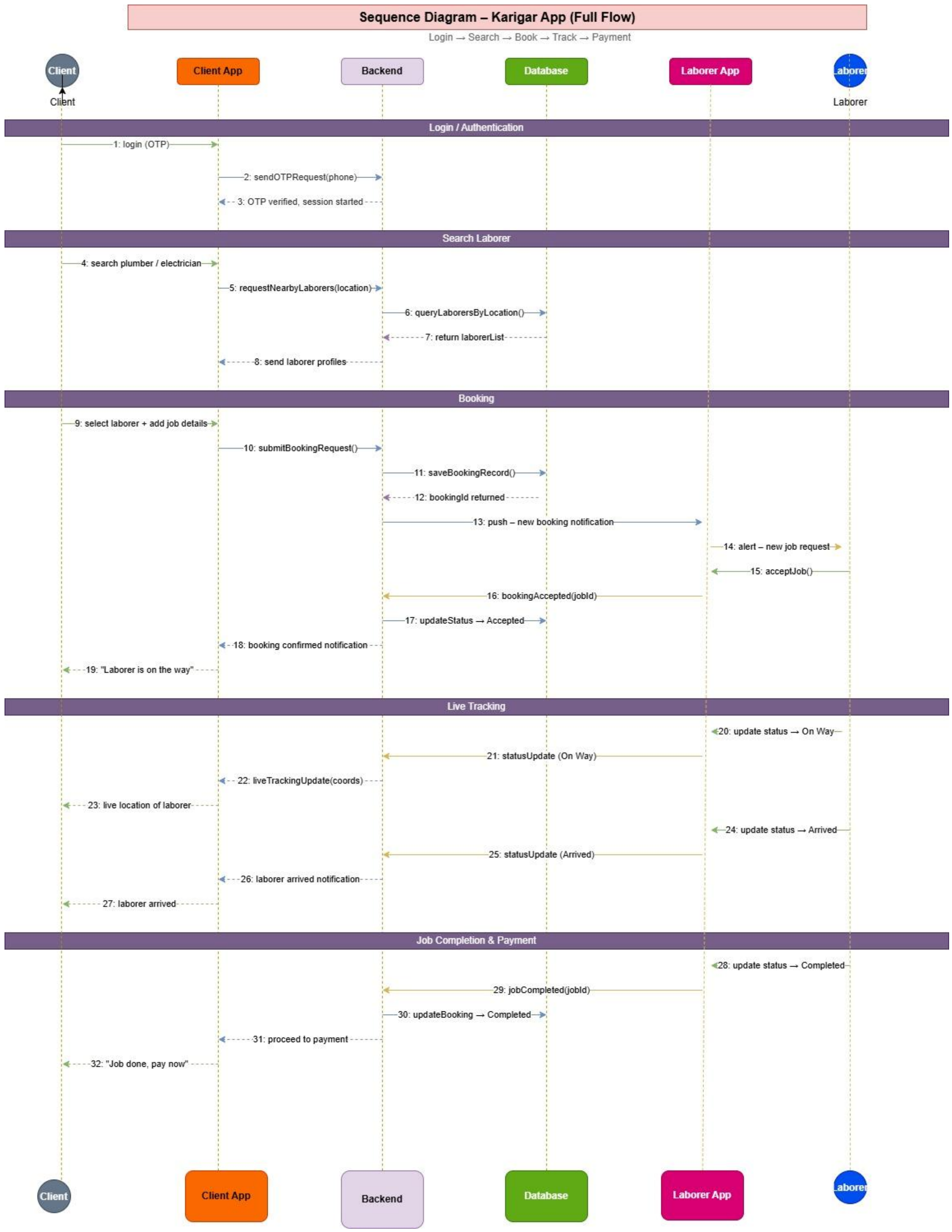
b. **Search Laborer:** The client searches for a plumber or electrician. The app forwards a location-based query to the backend, which retrieves matching laborer profiles from the database and returns them to the client.

c. **Booking:** The client selects a laborer and submits job details. The backend saves the booking record, returns a booking ID, and pushes a real-time notification to the Laborer App. The laborer accepts the job, triggering a status update to "Accepted" in the database, and the client receives a "Laborer is on the way" confirmation.

d. **Live Tracking:** The laborer updates their status to "On Way," which the backend relays to the Client App as live coordinate updates. Upon arrival, a further status update to "Arrived" is pushed and the client is notified accordingly.

e. **Job Completion & Payment:** The laborer marks the job as completed via the Laborer App. The backend updates the booking status to "Completed" in the database and prompts the client with a payment request, "Job done, pay now."

Figure 6. Karigar Sequence Diagram



i. System Sequence Diagram

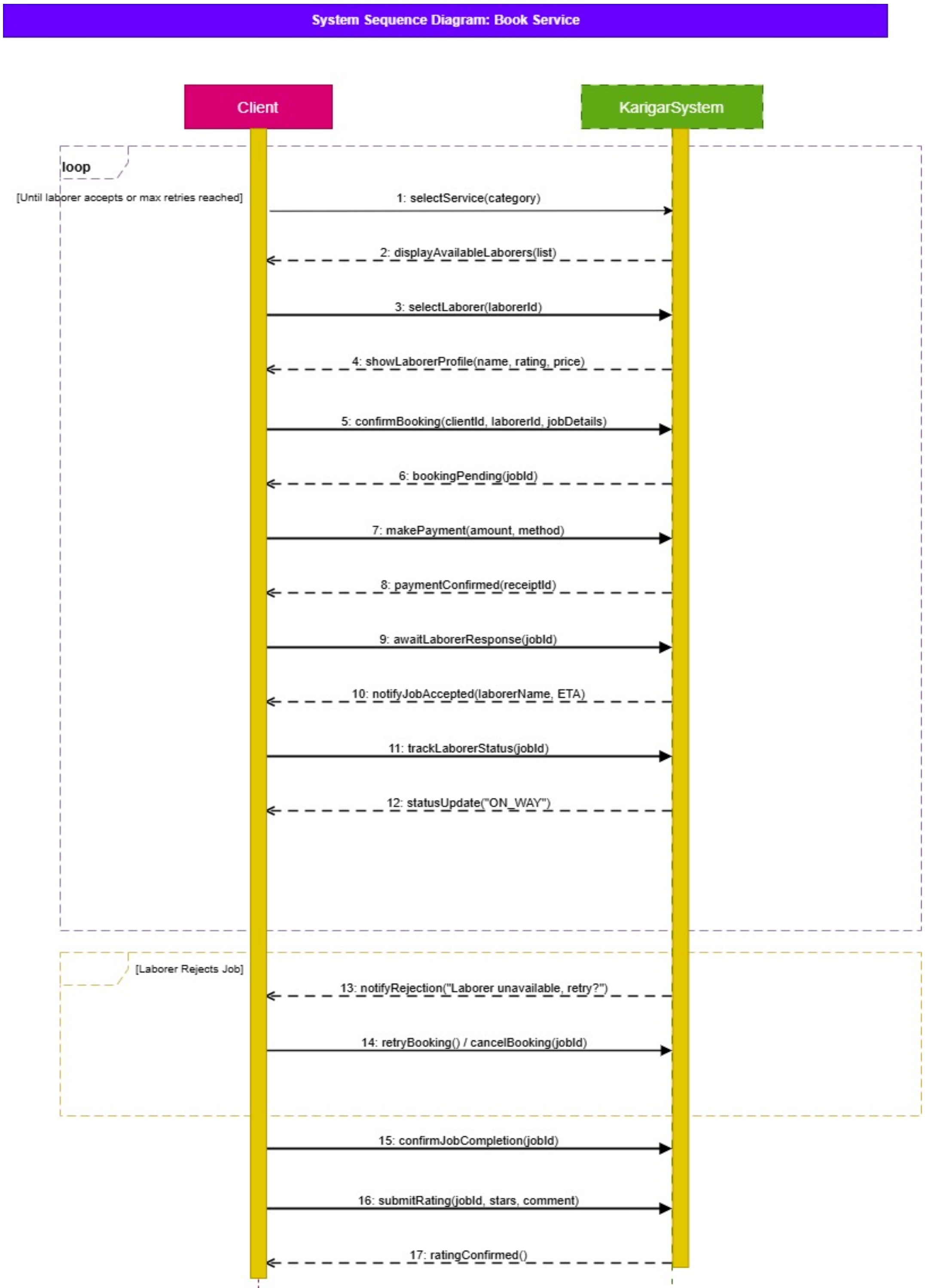
This diagram depicts the system-level interaction between the Client and the Karigar System as a black box, covering the complete book-service flow from selection through to post-job rating.

a. **Loop Frame:** Until Laborer Accepts or Max Retries Reached The client selects a service category, receives a list of available laborers, chooses one, and reviews their profile (name, rating, price). Upon confirming the booking with job details, the system returns a pending booking ID. The client then completes payment, receiving a receipt confirmation. The system awaits laborer acceptance and, upon approval, notifies the client with the laborer's name and ETA. Live status tracking begins, with the system responding with an ON_WAY update.

b. **Alternative Frame:** Laborer Rejects Job If the laborer is unavailable, the system notifies the client with a rejection prompt, offering the option to retry with another laborer or cancel the booking entirely, looping back if retried.

c. **Post-Job Flow:** Once the job is completed, the client confirms completion, submits a star rating with an optional comment, and receives a rating Confirmed acknowledgment from the system.

Figure 7. Karigar - System Sequence Diagram


System Sequence Diagram: Book Service
Client
KarigarSystem
loop
[Until laborer accepts or max retries reached]
1: selectService(category)
2: displayAvailableLaborers(list)
3: selectLaborer(laborerId)
4: showLaborerProfile(name, rating, price)
5: confirmBooking(clientId, laborerId, jobDetails)
6: bookingPending(jobId)
7: makePayment(amount, method)
8: paymentConfirmed(receiptId)
9: awaitLaborerResponse(jobId)
10: notifyJobAccepted(laborerName, ETA)
11: trackLaborerStatus(jobId)
12: statusUpdate("ON_WAY")
[Laborer Rejects Job]
13: notifyRejection("Laborer unavailable, retry?")
14: retryBooking() / cancelBooking(jobId)
15: confirmJobCompletion(jobId)
16: submitRating(jobId, stars, comment)
17: ratingConfirmed()

6. Conclusion

This report presents a comprehensive and structured analysis, design, and specification of the Karigar platform, developed to address the challenges associated with accessing reliable plumbing and electrical services in DHA Lahore. Through systematic requirements gathering, stakeholder analysis, feasibility evaluation, and detailed system modeling, the proposed solution demonstrates both practical viability and strong alignment with user needs.

The system emphasizes trust, efficiency, and usability by incorporating verified laborer onboarding, real-time service matching, transparent pricing, and secure payment mechanisms. Furthermore, the integration of user-centric features such as bilingual support, simplified workflows, and location-based services ensures accessibility for both clients and laborers.

From a technical and operational perspective, the platform is designed to be scalable, secure, and adaptable to future expansion beyond the initial deployment scope. The use of structured design artifacts, including use case diagrams, sequence diagrams, and detailed requirement specifications, ensures clarity and readiness for implementation.

In conclusion, the Karigar system represents a well-founded and feasible solution with the potential to significantly improve service accessibility, enhance user trust, and streamline the interaction between clients and skilled labor within a localized urban environment.

APPENDIX A: Interview Questionnaire

A.1 Clients (Residents)

Section A: Past Experience

Q1: "Think about the last time you had a plumbing or electrical issue at home. Walk me through exactly what you did, from problem to solution."

Probes:

- Who did you call? (Saved number? WhatsApp group? App?)
- How long did they take to arrive?
- What was the cost? How was it decided?
- Was the work done properly?

Q2: "How do you usually find plumbers or electricians? What works well and what doesn't?"

Probes:

- WhatsApp groups?
- Neighbor recommendations?
- Apps like Mahir?
- What problems do you face with your current method?

Section B: Trust & Verification

Q3: "When an app says 'verified laborer', what does that mean to you? What assurance do you actually get?"

Probes:

- Is police verification enough?
- How would you define a "trustworthy" laborer?

Q4: "Two laborers:

- Laborer A: 4.8 stars overall, 500 jobs
- Laborer B: 4.5 stars, but 20 jobs in your specific phase

Which one would you choose and why?"

Q5: "What would make you trust an app enough to use it regularly?"

Probes:

- Background checks?
- Money-back guarantee?
- Neighbor reviews?
- Fixed pricing?

Section C: Pricing and Emergencies

Q6: "It's 11 PM and your bathroom pipe bursts. What would you do? What would you expect from an app?"

Q7: "What's more important to you: lowest price or fastest response? Why?"

Q8: "If an app offered fixed rates (geyser repair Rs. 800, leaky tap Rs. 500), how would you feel?"

A.2 Commercial Clients (Office/Shops)

Section A: Past Experience

Q1: "How does your office maintenance currently work? Contract or on-demand?"

Q2: "For commercial use, how important is police verification compared to residential?"

Q3: "Would recurring maintenance visits (quarterly checks) be valuable?"

Q4: "After-hours service - how important for your business?"

A.3 Laborers (Plumbers and Electricians)

Section A: Current Work

Q1: "How do you usually get work? What's working well and what isn't?"

Probes:

- WhatsApp groups?
- Apps like Mahir?

- References?

Q2: "If you use apps like Mahir, what do you like and what frustrates you?"

Probes:

- Commission rate?
- Payment timing?
- Distance of jobs?
- Support when issues arise?

Section B: Incentives and Retention

Q3: "What would make you choose one app over another?"

Probes:

- Lower commission?
- Faster payment?
- Nearby jobs only?
- Training or benefits?

Q4: "Two apps:

- App A: 20% commission, no extras
- App B: 15% commission, uniform, ID card, monthly training

Which would you choose and why?"

Section C: Practical Needs

Q5: "How soon after completing a job should payment come?"

- Same day? 24 hours? Weekly?

Q6: "Would you prefer jobs only in your own phase (less travel) or jobs all over DHA (more jobs)?"

Q7: "Can you read English or should the app be in Urdu? Would voice notes help?"

Q8: "If the app offered free training on DHA-specific fixtures (Grohe, Paffoni), would you attend?"

APPENDIX B: Survey

- **Purpose:** Validate interview findings at scale
- **Target Participants:** DHA Residents (Phases 1-8)
- **Sample Size:** 100+ responses
- **Distribution:** WhatsApp groups, Facebook DHA groups
- **Duration:** 5 minutes
- **Tool:** Google Forms
- **Timeline:** 1 week

B.1 Survey Questions

Q1: Which DHA Phase do you live in?

- Phase 1-2
- Phase 3-4
- Phase 5-6
- Phase 7-8
- Adjacent block

Q2: In the last 6 months, how many times have you hired a plumber or electrician?

- 0
- 1-2
- 3-5
- 5+

Q3: How do USUALLY find laborers? (Select top 2)

- WhatsApp groups
- Neighbor recommendations
- Apps (Mahir, etc.)
- Saved numbers
- Facebook

Q4: What is your BIGGEST problem when hiring a laborer? (Select one)

- Can't find anyone available
- They arrive late
- Work quality is poor
- Price is uncertain/too high
- Don't trust them (safety)

Q5: How important is police verification to you?

- Not important
- Somewhat important
- Very important
- Must-have

Q6: Would you prefer a laborer from your own phase over someone from another phase?

- Strongly prefer
- Slightly prefer
- No preference

Q7: What response time do you expect for an EMERGENCY job?

- Within 15 minutes
- Within 30 minutes
- Within 1 hour
- Within 2 hours

Q8: Would you use an app specifically for DHA Lahore with verified plumbers/electricians?

- Definitely yes
- Probably yes
- Probably no
- Definitely no

Email/phone (for pilot testing - optional): ______

APPENDIX C: Competitor App Analysis

- **Purpose:** Identify gaps and strengths in existing solutions
- **Target Apps:** Mahir Company, Foran Connection
- **Method:** Download and use each app

- **Duration:** 3 days
- **Analysts:** 2 team members

C.1 Mahir Company

Table 49. Mahir Company

#	Aspect	*Observations*
1	**Registration Process**	What info required? How long?
2	**Booking Flow**	How many steps to book?
3	**Pricing Transparency**	Is price shown upfront?
4	**Location Filter**	Can you filter by area/phase?
5	**Payment Process**	Methods? When is laborer paid?
6	**Rating System**	Can anyone rate? Fake rating risk?
7	**Emergency Handling**	Is there emergency option?
8	**Laborer Experience**	Commission? Payment time?

C.2 Foran Connection

Table 50. Foran Connection

#	Aspect	*Observations*
1	**Registration Process**	What info required? How long?
2	**Booking Flow**	How many steps to book?
3	**Pricing Transparency**	Is price shown upfront?
4	**Location Filter**	Can you filter by area/phase?
5	**Payment Process**	Methods? When is laborer paid?

#	Aspect	***Observations***
6	**Rating System**	Can anyone rate? Fake rating risk?
7	**Emergency Handling**	Is there emergency option?
8	**Laborer Experience**	Commission? Payment time?